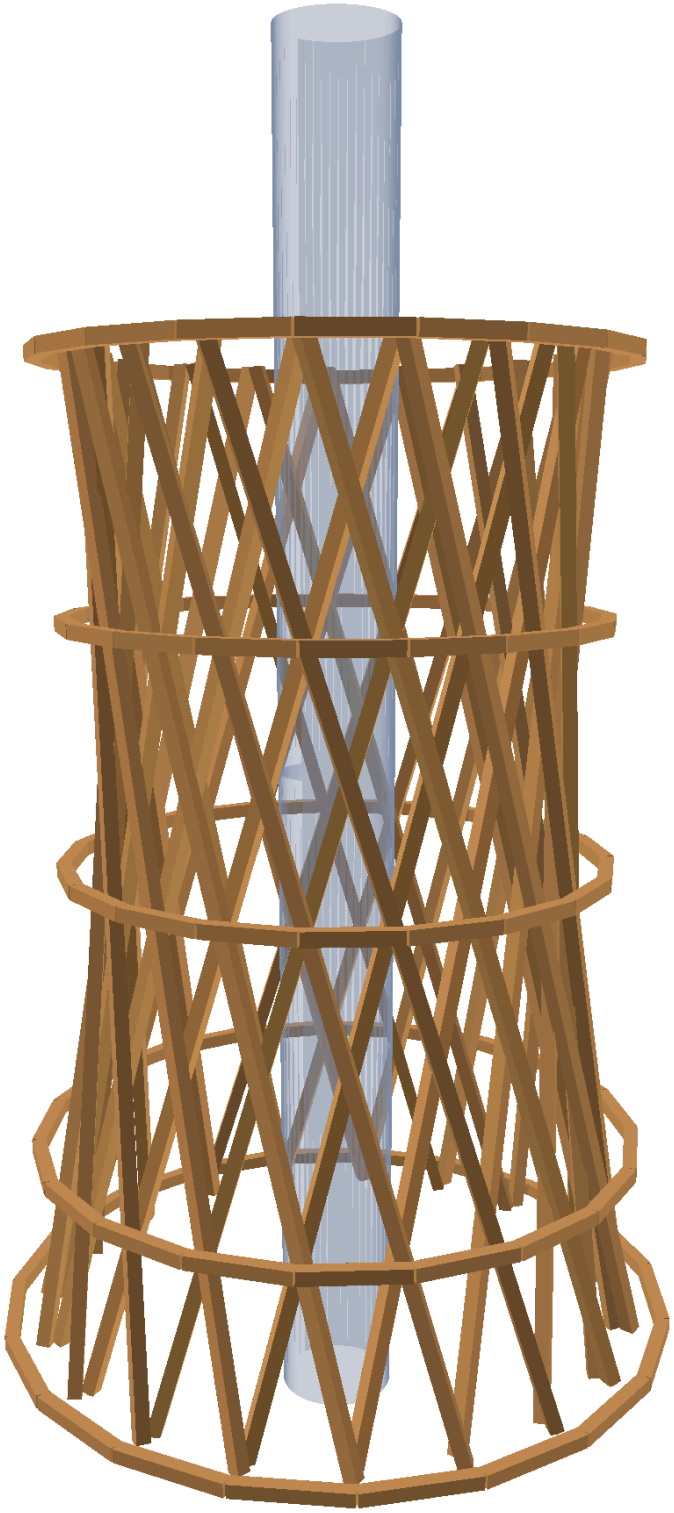
3×3 twisted diagrid

Assembly guide

Generated by Tower Designer. Brief: ≤ 20 g, 211 mm tall, fits over a 25 mm pole, built only from supplied 3 × 3 mm sticks (full size; only the joints are shaped).



Predicted failure load

83 kg

Predicted mass

19.24 g

wood 16.11 g + glue 3.13 g, including +3 % on every piece and joint

133 pieces · 399 glue joints · 13 sticks

(assuming 900 mm sticks)

Weakest point: Strip B3 crushes.
Gap to the pole: 30.6 mm all round.

Strength is based on **book values for medium balsa** because the stick species is unknown. Do the wood test first (section 5).

Contents

1. How to use this guide
2. How the tower works (2b: how the rings attach)
3. Safety
4. What you need
5. Before you start: test your wood and glue
6. Wood and grain direction
7. Glue plan
8. Printing the jig
9. Cutting the pieces
10. Build steps
11. Final checks
12. Test day
13. Troubleshooting and fixing mistakes
14. Appendix A: every piece
15. Appendix B: jig and model files

1. How to use this guide

- Every piece has an ID: A1–A19 and B1–B19 for the strips, R1-1, R3-1 … for the rings (ring number, then piece), BASE…, CAP… . Write it on the piece in pencil when you cut it.
- Every build step has a picture **and a 3D model** in the `steps/` folder. Open the models in Onshape (**+** → **Import**) and orbit round them if a picture isn't clear. In every model the **jig is blue**, pieces you have **already built are tan**, and **the pieces you add in that step are orange**.
- Every step lists the **jig parts** it uses by their engraved ID (dark tag, e.g. **SP2**) and file name in `jig/`.
- **Glue colour code**, the same in every picture, model, table and step: each glue point is a coloured mark.

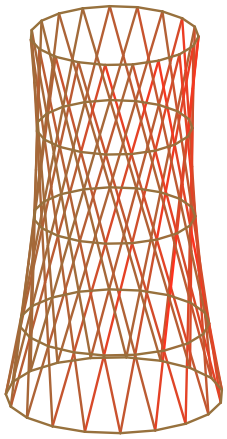
> Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

- ⚖ boxes are weigh-ins. Weigh the tower after each step once its glue is dry. If you are over, you are using too much glue.
- □ boxes are checks to tick before you move on. ⏸ boxes mean wait for the glue.

Words used in this guide

Strip	One of the long 220 mm sticks that run from the table to the top, leaning.
Set A / set B	The two sets of strips. A strips are the inner layer and lean one way; B strips are the outer layer and lean the other way. The pieces are identical until you letter them.
Letter face	The face you write A or B on. It always faces outwards on the tower.
Foot (F)	The end of a strip that stands on the table. You mark it F.
Crossing	Where a B strip passes over an A strip; they are glued face to face there.
Level	One of the heights where the strips cross (1 = lowest). Rings sit at some levels.
Tick	A pencil line on a strip, made with the marking gauge MG1, showing where a crossing goes.
Ring	A hoop of short pieces glued round the outside at one level.
Scarf	A long sloping end (1 in 3, about 9 mm) so two ring pieces overlap and glue on a big face.
Level cut	An angled end cut so a leaning strip stands flat on a level surface.
Lap joint	Two pieces glued face to face where one overlaps the other.
Pocket	A shallow V-shaped hole in the base plate where two strip feet stand.
Collar / segment	A printed ring that holds the strips at the right place; it comes in numbered segments so it can be taken out afterwards.
Notch	The slot in a collar segment that a strip presses into.
Mast / spacer	The rod up the middle of the jig, and the tubes stacked on it that set each collar's height. Each spacer has a disc on top that the collar segments rest on.
Grain	The direction of the wood fibres. In a stick it runs along its length.
End grain	The cut end of a stick. It soaks up glue and bonds poorly.
Double-glue	Balsa cement method: coat both faces, let it go tacky, coat again and press.
Cure	The time a glue needs to reach full strength.
Buckling	A slender piece bowing sideways under compression; the usual way thin sticks fail.

2. How the tower works



Colour = how hard each piece works at the predicted failure load (green light → red at its limit).

19 strips twist one way (**set A**, the inner layer) and 19 twist the other way (**set B**, the outer layer). Each strip turns through 66° on its way up. The base is Ø127 mm and the top Ø99 mm, measured to the strip centres, so the tower narrows like a cooling tower. Feet are 18.9° apart round the base. The strips cross at 6 levels: 43, 77, 105, 132, 157, 183 mm above the table. Rings sit at some of those levels. The printed jig sets every one of these positions for you.

Load path. The load plate presses on the top ring and the strip tops. Each strip carries its share straight down to the table. The strips lean in opposite directions, so where an A strip crosses a B strip they brace each other. That shortens the length that can buckle and turns the whole wall into a stiff lattice shell, like a woven basket.

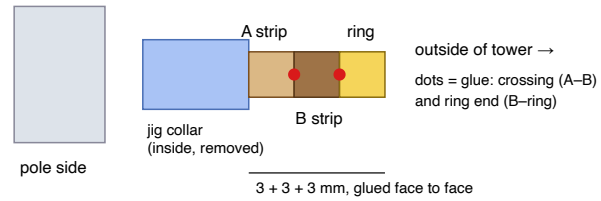
Why it narrows. A wider base resists tipping and buckling of the whole tower. The narrower top keeps the strips short and steep, so they stay in compression.

What the rings do. They hold the crossings at their levels so the lattice keeps its shape, and stop the strips spreading outwards at the base and squeezing inwards at the top.

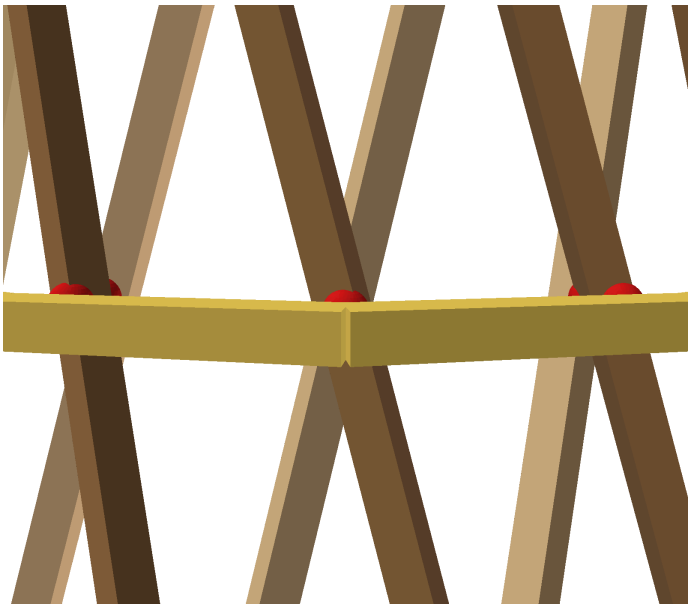
Why lap joints. Every joint is one face glued flat on another (side grain to side grain). That is far stronger than gluing onto a cut end.

How it is predicted to fail. Strip B3 crushes. The whole tower would buckle at about 178 kg, well above the failure load.

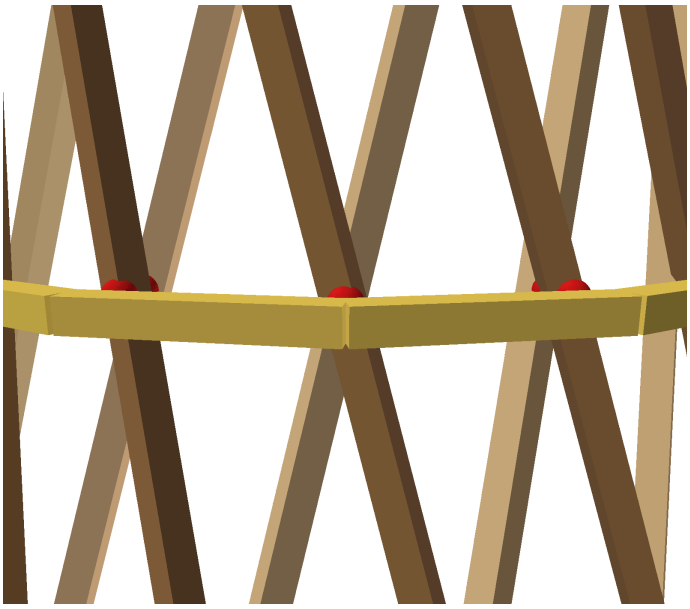
2b. How the rings attach, and why nothing hits the jig



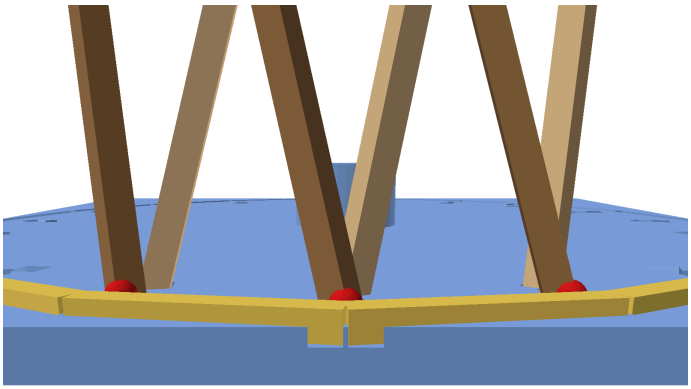
The tower is built in **three layers**, from the inside out: the A strips, then the B strips glued over them at every crossing, then the rings glued on the outside of the B strips. Each layer is one 3 mm stick thick, and every joint is a face glued flat on a face.



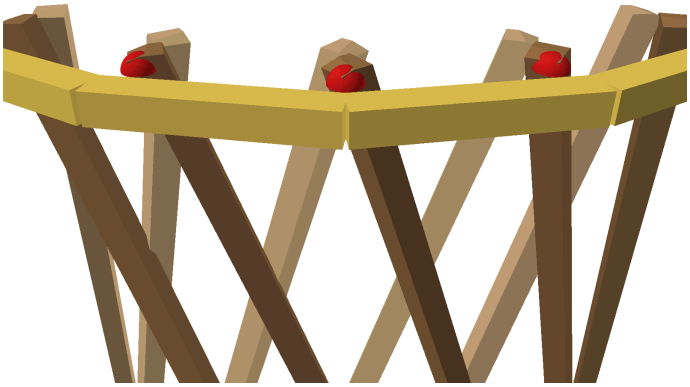
A crossing. The B strip lies over the A strip; glue goes where they touch (dot).



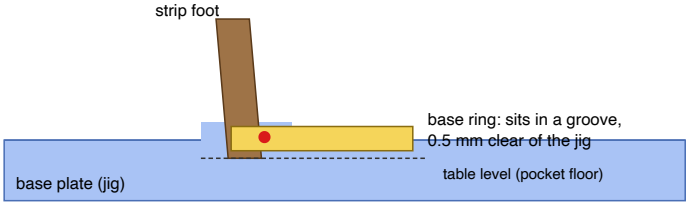
A ring at a crossing level. Ring pieces (gold) lie on the B strips. Neighbouring pieces **overlap each other on a 9 mm diagonal (scarf)** right over each B strip: one piece lies flat on the strip and the next lies on top of it. Glue goes on the strip patch and on the diagonal faces, so each ring is one continuous hoop.



The base ring in the jig. It lies on the outside of the feet, in a shallow groove in the base plate (blue) that keeps it 0.5 mm clear of the plastic.



The top ring. Its top is flush with the strip tops so the load plate presses on both.



Why the base ring doesn't hit the jig:

- The strip feet stand in 2 mm-deep pockets in the base plate. The base ring sits just above the feet on the outside, in its own groove, with 0.5 mm of air underneath, so it is never glued to or trapped by the plate.
- The collars hold the strips from the **inside**; every ring is on the **outside**. The lowest collar starts 19 mm up, well above the base ring.
- Collars sit between the ring levels, never at the same height.
- Checked in the 3D model: no jig part overlaps any piece of the tower anywhere.
- When you lift the finished tower off, the feet come straight up out of their pockets and the base ring out of its groove.

3. Safety

- Wear safety glasses when cutting, sanding, and crush-testing (wood splinters).
- Glue: work with a window open. Balsa cement gives off solvent fumes and is flammable: keep it away from flames and put the lid back on between joints. Acetone (nail-polish remover, for fixing mistakes) is flammable too.
- Razor saws and hobby knives: cut away from your fingers, on a cutting mat, with the stick held in the guide.
- Let the 3D printer cool before removing prints.

4. What you need

Materials

- **15 × 3 × 3 mm sticks**: 13 plus 2 spare (stick plan in section 9).
- UHU Hart balsa cement — hobby / model shops (not Bunnings)
- PLA filament for the jig, about 150 g.
- Packing tape, masking tape, baking paper.
- Acetone (nail-polish remover) and cotton buds, for undoing a wrong joint.

Tools

- Fine razor saw (or a new hobby-knife blade) and a cutting mat.
- 240-grit sandpaper glued flat to a board, plus a small sanding stick.
- Scale reading to 0.01 g. Ruler. Calipers if you have them.
- A small brush or toothpicks, scrap card, small spring pegs, sharp pencil, fine marker.
- 7 small cups or bags for sorting the pieces.
- Bambu printer (jig) and the app on the Mac.

5. Before you start: test your wood and glue

The prediction above uses book values and a standard glue allowance. Two quick tests make it match **your** sticks and **your** gluing. Both kits are in the `test kits/` folder of this build pack.

Wood test (`Wood test kit/`, about 30 min)

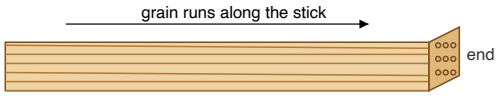
1. Weigh 5 whole sticks: this gives the density.
2. Crush 5 pieces 12 mm long, ends sanded square in `wood_test_squaring_block.stl`, between `wood_test_crush_pad.stl` pads on a 25 kg scale. Record each peak.
3. Press 3 straight 150 mm pieces in `wood_test_buckling_cap.stl` caps until they bow, and record the load.
4. Enter the results in the app (Wood and glue → Wood test…) and click Apply.

Glue test (`Glue test kit/`, a day for curing)

1. For each glue, weigh a 100 mm rod and 5 short strips together.
2. Glue the strips onto the rod in `glue_test_jig_base.stl` + `glue_test_jig_comb.stl`, exactly as you'll glue the tower.
3. When dry, weigh again. Enter the masses in the app (Glue test…) and click Apply.

After either test, **re-run the search and export the build pack again**. The design, cut list, and this guide may change.

6. Wood and grain direction



Side grain: long faces (good glue surface, strong)
End grain: the cut end (soaks up glue, weak bond)

The grain (the wood fibres) runs along the length of every 3 × 3 stick. Wood is strong and stiff along the grain: about **20–30× stiffer** than across it, and that is the direction every piece of this tower carries its load. **Never split or rip a stick lengthwise.** Only cut across it, so the grain always runs along each piece.



✔ grain parallel to the edges



✘ grain runs off the edge (snaps)

Pick straight-grained sticks. Hold each stick to the light and look at all four faces. Grain lines should run parallel to the edges. If they run off an edge ("run-out"), the stick will snap there. Use those sticks for short ring pieces only, cutting round the run-out.

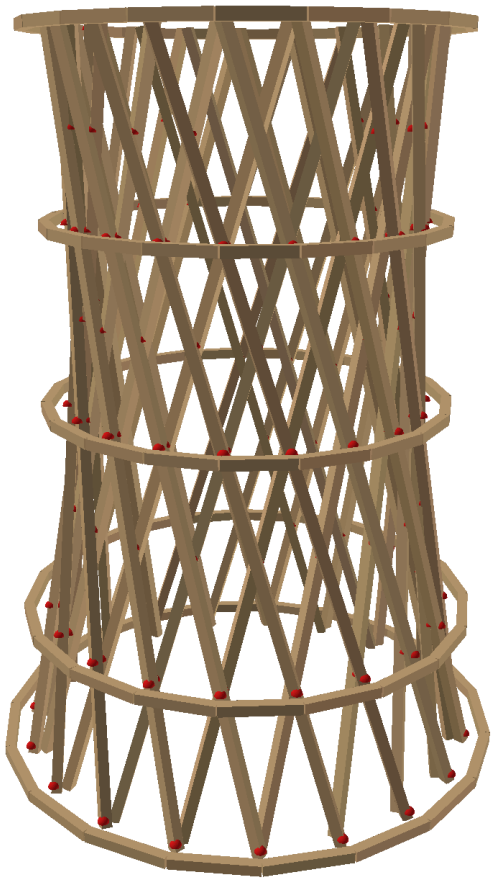
Sort the sticks before cutting:

1. Weigh every stick and write its mass on it. Mass ÷ volume (length × 9 mm²) gives the density; book balsa is about 160 kg/m³.
2. Rank them heaviest to lightest. Heavier (denser) sticks are stiffer and stronger.
3. Use the **heaviest 10 sticks** (sticks 1–10 in the stick plan) for the strips. They carry the load.
4. Use the lightest sticks for ring pieces, which carry little load.
5. Reject any stick that is bent, cracked, or much lighter than the others.

At the joints: glue side grain to side grain. A cut end (end grain) soaks up glue and bonds at only about 60 % of the strength, which is why this design laps every joint face-on-face. Keep sticks dry and flat; don't leave them in the sun or a damp room before building.

7. Glue plan

Glue dots: ● **Balsa cement**



Glue map: every glue point on the tower, coloured by the glue to use.
Each step's picture and 3D model shows the same dots.

Every joint uses Balsa cement. "Apply" is roughly what to put on; "dries to" is the most that should stay in the joint. A pinhead dab is about 2–4 mg.

Where	Joint	Glue	Count	Apply (wet)	Dries to
Base ring → Base ring	scarf	Balsa cement	19	≈ 42 mg	≤ 13 mg
Base ring → the strips (lapped across the outside)	scarf end	Balsa cement	38	≈ 13 mg	≤ 4 mg
Crossings (B strip over A strip)	lap	Balsa cement	114	≈ 35 mg	≤ 11 mg
Ring 1 → Ring 1	scarf	Balsa cement	19	≈ 42 mg	≤ 13 mg
Ring 1 → the strips (lapped across the outside)	scarf end	Balsa cement	38	≈ 13 mg	≤ 4 mg
Ring 3 → Ring 3	scarf	Balsa cement	19	≈ 42 mg	≤ 13 mg
Ring 3 → the strips (lapped across the outside)	scarf end	Balsa cement	38	≈ 13 mg	≤ 4 mg
Ring 5 → Ring 5	scarf	Balsa cement	19	≈ 42 mg	≤ 13 mg
Ring 5 → the strips (lapped across the outside)	scarf end	Balsa cement	38	≈ 13 mg	≤ 4 mg
Top cap → Top cap	scarf	Balsa cement	19	≈ 42 mg	≤ 13 mg
Top cap → the strips (lapped across the outside)	scarf end	Balsa cement	38	≈ 13 mg	≤ 4 mg

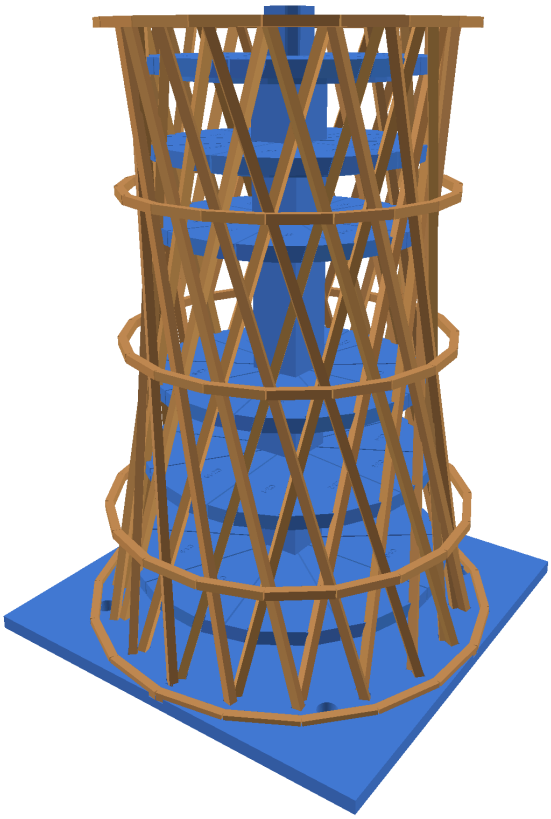
How to use each glue

Glue	Method	Cure	In this tower
Balsa cement UHU Hart balsa cement — hobby / model shops (not Bunnings)	Double-glue. Brush a thin coat on both faces, wait 5–10 min until it is tacky-dry (it soaks into the wood), add a second thin coat to one face and press together for 30 s. Leave 12 h before loading.	Tacky in 1–2 min, full strength after a few hours	399 joints, 3.13 g dry

Preparing a joint: make sure the two faces touch flat with no light showing between them. If they don't, touch the contact area (only) once on the sandpaper. Blow off dust before gluing. Don't fill gaps with extra glue; fix the fit instead.

If you'd rather not use balsa cement: With PVA on every joint, this tower would weigh 25.65 g and hold 83 kg — over the limit. To switch, set Glue to PVA only in the app, re-run the search, and export again.

8. Printing the jig



All 73 files are in `jig/` as **STL** (for Bambu Studio) and **STEP** (for Onshape). `jig/jig_assembly.step` shows the whole jig assembled.

Print settings (PLA): 0.2 mm layers, 3 walls, 15 % infill, no supports, brim off. Print each part in the orientation in the table below: collar segments with the engraved ID facing up (the notches lean, so upside down is wrong), spacers disc-down, the mast lying flat.

Print in this order (rough times on a Bambu A1/P1):

Plate	What	Parts	About
1	Test first	FT	≈ 5 g, 0.4 h
2	Preparation jigs	CG1, MG1, SG, SJ1	≈ 131 g, 11.9 h
3	Base plate	BP	≈ 62 g, 5.7 h
4	Mast and spacers	SP1, SP2, SP3, SP5, SP6	≈ 39 g, 3.5 h
5	Mast and spacers	M, SP4	≈ 26 g, 2.3 h
6	Collar C1	C1-1, C1-2, C1-3, C1-4, C1-5, C1-6, C1-7, C1-8, C1-9, C1-10	≈ 29 g, 2.6 h
7	Collar C2	C2-1, C2-2, C2-3, C2-4, C2-5, C2-6, C2-7, C2-8, C2-9, C2-10	≈ 23 g, 2.1 h
8	Collar C3	C3-1, C3-2, C3-3, C3-4, C3-5, C3-6, C3-7, C3-8, C3-9, C3-10	≈ 19 g, 1.7 h
9	Collar C4	C4-1, C4-2, C4-3, C4-4, C4-5, C4-6, C4-7, C4-8, C4-9, C4-10	≈ 17 g, 1.5 h
10	Collar C5	C5-1, C5-2, C5-3, C5-4, C5-5, C5-6, C5-7, C5-8, C5-9, C5-10	≈ 17 g, 1.5 h
11	Collar C6	C6-1, C6-2, C6-3, C6-4, C6-5, C6-6, C6-7, C6-8, C6-9, C6-10	≈ 18 g, 1.7 h

After printing:

1. Print **FT** first and check your sticks fit (Step 0, item 1) before printing anything else.
2. Remove any stringing from the notches and slots with a knife.
3. Check every part's engraved ID is readable (C1-1, C1-2 …, SP1 …); go over faint ones with a fine marker.
4. Stack the jig dry on the mast (Step 1) once before you start, to check every segment sits on its number.

Every part has its **ID engraved** on it (the bold tag below). The guide uses these IDs everywhere.

ID	File	Qty	Used in	Print orientation
BP	`base_plate.stl`	1	steps 1–8	flat face down
C1-1 … -10	`collar_1_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
C2-1 … -10	`collar_2_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
C3-1 … -10	`collar_3_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
C4-1 … -10	`collar_4_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
C5-1 … -10	`collar_5_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
C6-1 … -10	`collar_6_seg1 … seg10.stl`	10	steps 1–8	engraved ID facing up
CG1	`cutting_guide_1.stl`	1	Step 0 (preparing parts)	flat face down
FT	`fit_test_coupon.stl`	1	Step 0 (preparing parts)	flat face down
M	`mast.stl`	1	steps 1–8	lying flat on the bed
MG1	`marking_gauge.stl`	1	Step 0 (preparing parts)	flat face down
SG	`scarf_sanding_guide.stl`	1	Step 0 (preparing parts)	wide flat bottom down
SJ1	`end_sanding_jig_17deg.stl`	1	Step 0 (preparing parts)	wide flat bottom down
SP1	`spacer_1.stl`	1	steps 1–8	disc down (upside down)
SP2	`spacer_2.stl`	1	steps 1–8	disc down (upside down)
SP3	`spacer_3.stl`	1	steps 1–8	disc down (upside down)
SP4	`spacer_4.stl`	1	steps 1–8	disc down (upside down)
SP5	`spacer_5.stl`	1	steps 1–8	disc down (upside down)
SP6	`spacer_6.stl`	1	steps 1–8	disc down (upside down)

Stack on the mast, bottom to top (heights above the table):

Part	Sits at
BP base_plate.stl boss	up to 14.0 mm
SP1 spacer_1.stl (4.8 mm tall, disc on top)	14.0 → 18.8 mm
C1-1 … C1-10 collar_1_seg1 … seg10.stl, each on its number	18.8 → 23.8 mm
SP2 spacer_2.stl (33.3 mm tall, disc on top)	23.8 → 57.1 mm
C2-1 … C2-10 collar_2_seg1 … seg10.stl, each on its number	57.1 → 62.1 mm

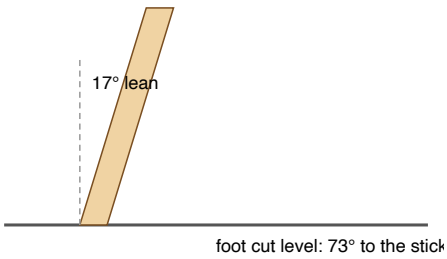
SP3 spacer_3.stl (26.4 mm tall, disc on top)	62.1 → 88.5 mm
C3-1 … C3-10 collar_3_seg1 … seg10.stl, each on its number	88.5 → 93.5 mm
SP4 spacer_4.stl (48.5 mm tall, disc on top)	93.5 → 142.0 mm
C4-1 … C4-10 collar_4_seg1 … seg10.stl, each on its number	142.0 → 147.0 mm
SP5 spacer_5.stl (20.6 mm tall, disc on top)	147.0 → 167.6 mm
C5-1 … C5-10 collar_5_seg1 … seg10.stl, each on its number	167.6 → 172.6 mm
SP6 spacer_6.stl (21.9 mm tall, disc on top)	172.6 → 194.5 mm
C6-1 … C6-10 collar_6_seg1 … seg10.stl, each on its number	194.5 → 199.5 mm

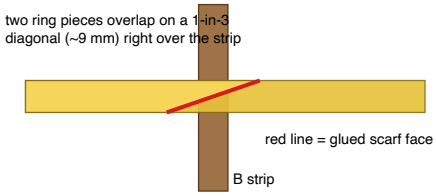
9. Cutting the pieces

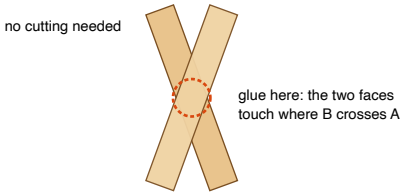
Every piece is a full 3 × 3 stick cut to length; only the ends are shaped. The cutting guide **CG1** has one channel per piece, numbered and labelled with its length. Each channel is open at the left, has a saw slot near the left end and a **stop** exactly one piece-length to the right of the slot. You never measure.

1. Tape the guide to the bench. Pick the channel for the piece (table below).
2. **Square up the start.** Lay the stick in the channel from the left so its end pokes about 5 mm past the slot, and saw through the slot. Throw the offcut away.
3. **Cut a piece.** Slide the stick to the right until its fresh end touches the stop. Hold it there with a finger well away from the slot, and saw gently through the slot. The piece between the slot and the stop is exactly the right length, and the end left on the stick is already cut for the next piece.
4. Repeat step 3 along the stick. **Don't turn the stick over between cuts**, so the angled strip ends stay parallel.
5. Write the piece ID on it in pencil and drop it in its cup. Strips also get their letter and F (Step 0).

Piece	Qty	Length	End 1	End 2	Cut with	Sand ends with
Strip A A1…A19	19	**220.2 mm**	level cut, ~17° off square	level cut, ~17° off square	**CG1** ch 1	**SJ1**
Strip B B1…B19	19	**220.2 mm**	level cut, ~17° off square	level cut, ~17° off square	**CG1** ch 1	**SJ1**
Base ring BASE1…BASE19	19	**30.0 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	**CG1** ch 2	**SG**
Ring 1 R1-1…R1-19	19	**27.3 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	**CG1** ch 3	**SG**
Top cap CAP1…CAP19	19	**25.3 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	**CG1** ch 4	**SG**
Ring 3 R3-1…R3-19	19	**24.6 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	**CG1** ch 5	**SG**
Ring 5 R5-1…R5-19	19	**24.1 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	**CG1** ch 6	**SG**







- **Level cuts (strip ends):** the strips lean, so both ends are cut at the same angle and parallel to each other. The cutting guide gets the angle close. The sanding jig **SJ1** then makes it exact, including the slight inward tilt of the tower.
- **Ring ends:** Neighbouring pieces **overlap each other on a 9 mm diagonal (scarf)** right over each B strip: one piece lies flat on the strip and the next lies on top of it. Glue goes on the strip patch and on the diagonal faces, so each ring is one continuous hoop. Cut the ring pieces square, then sand each scarf in **SG** (Step 0): end 1 in groove O UP, end 2 in groove O DOWN, so the overlaps stack.
- **Crossings:** no cutting at all. The faces touch where B crosses A; you only glue there.
- Cut 2 or 3 spares of each ring length. Ring pieces are tiny and easy to lose or break.

Stick plan (which pieces come from which stick, in mm, allowing 5 mm to square up the start and 2 mm per saw cut). Cut the strips from the heaviest sticks:

- Stick 1: A5 (220.2), A6 (220.2), A8 (220.2), A9 (220.2)
- Stick 2: A10 (220.2), A11 (220.2), A13 (220.2), B3 (220.2)
- Stick 3: B4 (220.2), B5 (220.2), B7 (220.2), B8 (220.2)
- Stick 4: B10 (220.2), B11 (220.2), B13 (220.2), B14 (220.2)
- Stick 5: B18 (220.2), B19 (220.2), A1 (220.2), A2 (220.2)
- Stick 6: A3 (220.2), A4 (220.2), A7 (220.2), A12 (220.2)
- Stick 7: A14 (220.2), A15 (220.2), A16 (220.2), A17 (220.2)
- Stick 8: A18 (220.2), A19 (220.2), B1 (220.2), B2 (220.2)
- Stick 9: B6 (220.2), B9 (220.2), B12 (220.2), B15 (220.2)
- Stick 10: B16 (220.2), B17 (220.2), BASE17 (30.0), BASE13 (30.0), BASE12 (30.0), BASE19 (30.0), BASE2 (30.0), BASE6 (30.0), BASE10 (30.0), BASE15 (30.0), BASE16 (30.0), BASE1 (30.0), BASE4 (30.0), BASE5 (30.0), BASE8 (30.0), BASE9 (30.0)
- Stick 11: BASE3 (30.0), BASE7 (30.0), BASE11 (30.0), BASE18 (30.0), BASE14 (30.0), R1-17 (27.3), R1-13 (27.3), R1-12 (27.3), R1-19 (27.3), R1-8 (27.3), R1-2 (27.3), R1-6 (27.3), R1-15 (27.3), R1-4 (27.3), R1-5 (27.3), R1-9 (27.3), R1-10 (27.3), R1-16 (27.3), R1-1 (27.3), R1-3 (27.3), R1-7 (27.3), R1-11 (27.3), R1-14 (27.3), R1-18 (27.3), CAP17 (25.3), CAP12 (25.3), CAP19 (25.3), CAP10 (25.3), CAP13 (25.3), CAP1 (25.3)
- Stick 12: CAP3 (25.3), CAP4 (25.3), CAP5 (25.3), CAP7 (25.3), CAP8 (25.3), CAP16 (25.3), CAP6 (25.3), CAP15 (25.3), CAP9 (25.3), CAP2 (25.3), CAP11 (25.3), CAP18 (25.3), CAP14 (25.3), R3-17 (24.6), R3-12 (24.6), R3-19 (24.6), R3-13 (24.6), R3-5 (24.6), R3-6 (24.6), R3-8 (24.6), R3-10 (24.6), R3-16 (24.6), R3-3 (24.6), R3-15 (24.6), R3-1 (24.6), R3-2 (24.6), R3-4 (24.6), R3-9 (24.6), R3-7 (24.6), R3-11 (24.6), R3-14 (24.6), R3-18 (24.6), R5-17 (24.1)

- Stick 13: R5-12 (24.1), R5-19 (24.1), R5-13 (24.1), R5-8 (24.1), R5-15 (24.1), R5-1 (24.1), R5-4 (24.1), R5-6 (24.1), R5-10 (24.1), R5-2 (24.1), R5-3 (24.1), R5-5 (24.1), R5-7 (24.1), R5-9 (24.1), R5-16 (24.1), R5-11 (24.1), R5-14 (24.1), R5-18 (24.1)

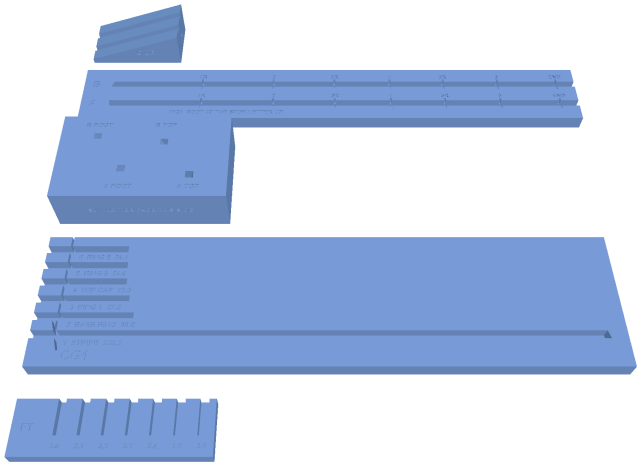
10. Build steps

Before Step 1: all pieces cut and labelled, jig printed and test-fitted, glues and tools laid out, a clear flat bench.

A sensible plan (each session 1–2 hours):

Session	Do	Then wait
1	Step 0: test fit, cut, letter, mark and sand every piece	—
2	Steps 1–2 (jig and A strips, no glue), then the first half of Step 3	1 h between batches
3	The rest of Step 3 (crossings)	12 h
4	Base ring and the first ring	1 h
5	The other rings and the top ring	24 h
6	Step 9: remove the jig, finish, weigh	—

Step 0. Preparing the parts



The preparation jigs, each with its ID engraved.

Jigs for this step:

- FT `fit_test_coupon.stl` : finds the slot size your sticks need. Print it first.
- CG1 `cutting_guide_1.stl` : one channel per piece, each with a saw slot and a stop one piece-length away. Channel numbers and lengths are engraved beside each channel.
- MG1 `marking_gauge.stl` : marks where every crossing and ring goes on a strip, and checks its length.
- SJ1 `end_sanding_jig_17deg.stl` : Strip ends, about 17° off square plus the inward tilt. Four slots: A foot, A top, B foot, B top. Letter face to the front.
- SG `scarf_sanding_guide.stl` : makes the 1-in-3 scarf on every ring end. Two grooves: O UP and O DOWN.

Also: razor saw, 240-grit sandpaper glued flat to a board, sharp pencil, 0.01 g scale, 7 small cups or bags.

Everything you need to make (133 pieces from 13 sticks). Tick each row when its pieces are cut, sanded and labelled:

Piece	Qty	Length	End 1	End 2	Cut with	Sand with	From stick	Group mass
☐ **Strip A** A1…A19	**19**	**220.2 mm**	level cut, ~17° off square	level cut, ~17° off square	CG1 ch 1	SJ1	1–8	≤ 6.21 g
☐ **Strip B** B1…B19	**19**	**220.2 mm**	level cut, ~17° off square	level cut, ~17° off square	CG1 ch 1	SJ1	2–10	≤ 6.21 g
☐ **Base ring** BASE1…BASE19	**19**	**30.0 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	CG1 ch 2	SG	10–11	≤ 0.84 g
☐ **Ring 1** R1-1…R1-19	**19**	**27.3 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	CG1 ch 3	SG	11	≤ 0.77 g
☐ **Top cap** CAP1…CAP19	**19**	**25.3 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	CG1 ch 4	SG	11–12	≤ 0.71 g
☐ **Ring 3** R3-1…R3-19	**19**	**24.6 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	CG1 ch 5	SG	12	≤ 0.69 g
☐ **Ring 5** R5-1…R5-19	**19**	**24.1 mm**	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	CG1 ch 6	SG	12–13	≤ 0.68 g

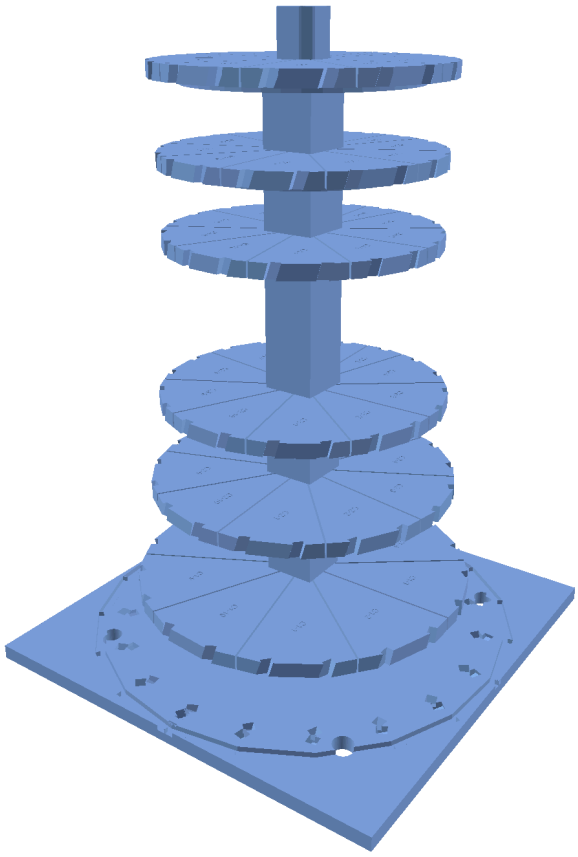
Do this:

1. **Check the fit.** Slide a stick into each slot of FT . Find the smallest slot it slides into with light friction. If that isn't 3.3, set **Jig slot size** in the app to that number and export the build pack again before printing the other jigs.
2. **Sort the sticks.** Weigh each stick and number them 1–13 in pencil, heaviest first. The stick plan in section 9 puts the strips on sticks 1–10.
3. **Label 7 cups:** Strip A (A1–A19, 19); Strip B (B1–B19, 19); Base ring (BASE1–BASE19, 19); Ring 1 (R1-1–R1-19, 19); Top cap (CAP1–CAP19, 19); Ring 3 (R3-1–R3-19, 19); Ring 5 (R5-1–R5-19, 19). Every piece goes straight into its cup.
4. **Cut to length** in CG1 (full method in section 9). Square up the stick's end at the slot first. Then for each piece: push the stick along until its end touches the stop, and saw at the slot. No measuring: the stop sets the length.
5. **Letter the strips as they come off the guide.** Each strip lies in the channel with its foot at the left (the slot end).
   - Strip A: write **A** and its number on the **top** face near the left end, then **F** at that end.
   - Strip B: **roll it over** (keep the left end on the left), then write **B** and its number on the new top face near the left end, and **F** at that end.

   The lettered face is the **letter face**: it faces outwards on the tower.
6. **Mark the crossings** in MG1 . Lay each strip in its channel (A or B), letter face up, foot against the angled stop. Pencil a tick through every notch onto the letter face and carry it round onto both side faces. Copy the number beside the notch (1, 2, 3R …; R means a ring goes there). The strip's top end should line up with the END notch, within 1 mm.
7. **Sand the strip ends in SJ1 .** Put the jig on the sandpaper board, letter face of the strip towards the side marked LETTER FACE THIS SIDE. Push the foot (F) end down the slot labelled with its letter and FOOT (for example A FOOT) until it touches the sandpaper. Pinch the strip to the jig so it can't slide, and rub in small strokes. Stop as soon as the plastic all round the slot shows sanding marks: then the end is flush and exact. Do the top end the same way in the matching TOP slot. Check: the strip stands on the table at its lean without rocking. Never push the strip further down while sanding; that only makes it shorter.
8. **Scarf the ring ends in SG .** Pencil an **O** on one face of each ring piece (its outer face). End 1: lay the piece in groove **O UP** (O face upwards) with its end 1 mm past the thin front edge, press it down with a finger and rub the guide on the sandpaper board until the plastic either side starts to scuff. End 2: the same in groove **O DOWN** (O face downwards). Pencil a small 2 next to end 2. Check: two pieces laid together, O faces up, overlap on their slopes with no light between them.
9. **Count and check.** Count each cup against the table (cut 2–3 spare ring pieces of each size). Weigh each cup: it should be no more than the group mass shown, or swap in pieces from lighter sticks.

⚖ All pieces together should weigh no more than **16.11 g** before any glue.

Step 1. Set up the jig



3D model of this step: `steps/step_01_jig.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:

- **BP** `base_plate.stl`
- **M** `mast.stl`
- **SP1** `spacer_1.stl`
- **SP2** `spacer_2.stl`
- **SP3** `spacer_3.stl`
- **SP4** `spacer_4.stl`
- **SP5** `spacer_5.stl`
- **SP6** `spacer_6.stl`
- **C1-1 … C1-10** `collar_1_seg1.stl` … `collar_1_seg10.stl` (10 pieces)
- **C2-1 … C2-10** `collar_2_seg1.stl` … `collar_2_seg10.stl` (10 pieces)
- **C3-1 … C3-10** `collar_3_seg1.stl` … `collar_3_seg10.stl` (10 pieces)
- **C4-1 … C4-10** `collar_4_seg1.stl` … `collar_4_seg10.stl` (10 pieces)
- **C5-1 … C5-10** `collar_5_seg1.stl` … `collar_5_seg10.stl` (10 pieces)
- **C6-1 … C6-10** `collar_6_seg1.stl` … `collar_6_seg10.stl` (10 pieces)

Pieces added: none

Do this:

1. Lay a sheet of baking paper on a flat bench and tape the base plate **BP** down on it, pockets facing up.
2. Push the mast **M** into the socket in the middle of the boss. One corner of the mast is cut off, and so is one corner of the socket: line them up and it goes in only one way. Push it down until it stops. If it wobbles, wrap one layer of tape round its foot and push it in again.
3. Build the stack on the mast from the bottom up, using the stack table in section 8: **SP1**, then the segments of collar **C1**, then **SP2**, then **C2**, and so on up to **C6**. Each spacer has the same cut corner in its hole, so it only goes on one way, with its disc on top.
4. Before you cover a spacer's disc, read the numbers printed on it: each number is where that collar segment goes. Put segment **C1-1** on the **1**, **C1-2** on the **2**, and so on. Engraved ID on top, notches facing outwards, inner end touching the mast. The numbers run anticlockwise (seen from above) from the **A1 B1** mark on the base plate.
5. Put packing tape over the base-plate pockets and the base-ring groove, so glue can't bond the tower to the plate. (No glue goes near the collars, so they don't need tape.)
6. Cut 38 tabs of masking tape, about 6 mm × 30 mm, and stick them along the edge of the bench ready for Step 2.

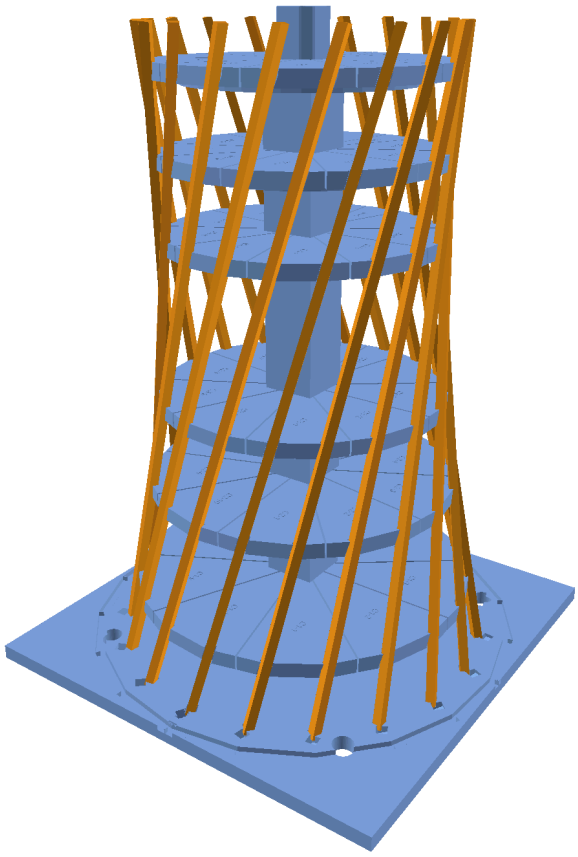
Check:

- □ The mast is upright: sight it against a door frame from two sides.
- □ Every collar is complete and flat, and each segment sits on its own number.
- □ Every segment's ID is on top, and the IDs read C1-…, C2-… from the bottom up.

Common mistakes:

- Stacking the spacers in the wrong order puts the collars at the wrong heights, so the strips won't follow their notches. The disc on each spacer says which collar goes on it ("under C3").

Step 2. Inner strips (set A)



3D model of this step: `steps/step_02_strips_a.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

Pieces added: 19 pieces: A1, A2, A3, A4, A5, A6, A7, A8, A9, A10, A11, A12, A13, A14, A15, A16, A17, A18, A19

Do this:

1. Take the A strips (A1–A19). Each has a letter **A** on its outer face and a foot end marked **F** (Step 0).
2. Seen from outside, the A strips lean to the **right** as they go up (the B strips lean the other way).
3. Each pocket in the base plate is a V: two strips share each foot. **A1** goes in the pocket at the **A1 B1** mark, **A2** in the next pocket to the right (anticlockwise seen from above), and so on round.
4. Stand the strip with its foot in the **inner** arm of the V, letter face outwards, lean it to the right, and press it into the notch of every collar it passes, from C1 up to C6.
5. Hold it with a masking-tape tab at collars C2, C5: tape over the strip with the ends pressed onto the top and bottom of the collar. The tabs sit between the B strips' paths, so the B strips never lie on them.
6. Check the angled foot sits flat in its pocket and the top end is 211 mm above the table (measure one).
7. No glue yet. This step only sets out the inner layer.

Check:

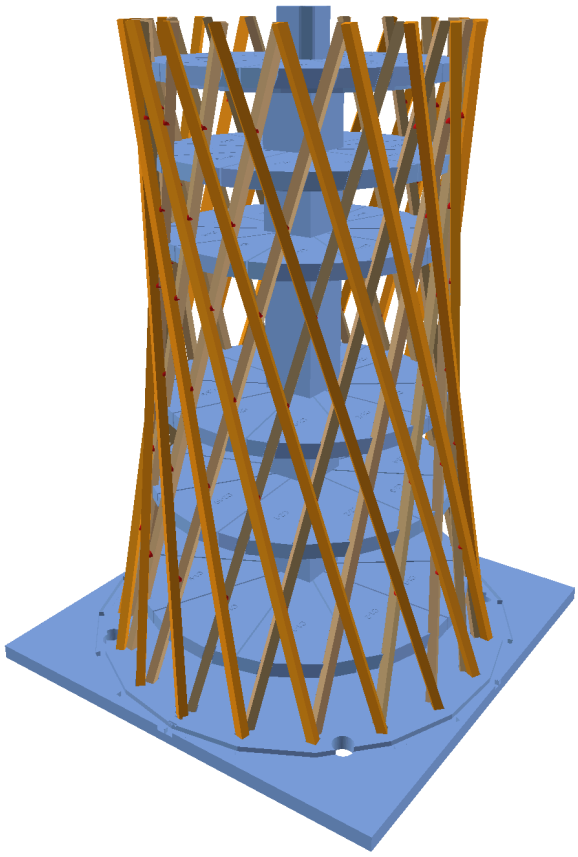
- □ All 19 A strips are in pockets and notches, letter faces outwards.
- □ The strips spiral evenly: the gaps between neighbours look the same all the way up.
- □ The pencil ticks on every strip are at the same heights all round.

Common mistakes:

- A strip in the wrong pocket makes every crossing above it miss. Count the pockets from the mark.

⚖ Weigh these pieces before going on: together they should be no more than **6.21 g**. If they are heavier, swap in lighter sticks now.

Step 3. Outer strips (set B) and the crossings



3D model of this step: `steps/step_03_strips_b.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

Pieces added: 19 pieces: B1, B2, B3, B4, B5, B6, B7, B8, B9, B10, B11, B12, B13, B14, B15, B16, B17, B18, B19

Glue dots: ● **Balsa cement**

Glue in this step:

- 114 × lap with **Balsa cement**: apply ≈ 35 mg each, dries to ≤ 11 mg

Do this:

1. Take the B strips (B1–B19). They lean to the **left** going up, seen from outside, and sit in the **outer** arm of each V pocket, letter face outwards. B1 shares A1's pocket.
2. **Dry fit first.** Stand B1 in its pocket and lay it over the A strips. Each pencil tick on B1 should sit right over the crossing with an A strip (the A strip's tick at the same height). Take it out again.
3. **Glue in batches of three strips.** Balsa cement needs two coats. Brush a thin coat on each B strip's **inner** face (opposite the letter) at its six ticks, and on the A strips' outer face at the six crossings each B strip will touch. Wait 5–10 min until it's tacky.
4. Add a second thin coat on the A side, stand the B strip in its pocket and lay it on. Press each crossing for 30 s, working from the bottom up. If a crossing springs apart, clip a small spring peg across it (one jaw inside the A strip, one outside the B strip) until it grabs.
5. Go round to the right in order: B1–B3, then B4–B6, and so on. That's 114 crossings, about 2 hours. Split it over two sessions if you like; stop after a complete batch.
6. Wipe off any glue that squeezes out. It adds mass and no strength.

Check:

- □ Every crossing has glue and the faces touch fully.
- □ B strips sit tight against A; no gaps over 0.5 mm.
- □ All the masking-tape tabs are still on the outside of the A strips and free of glue.

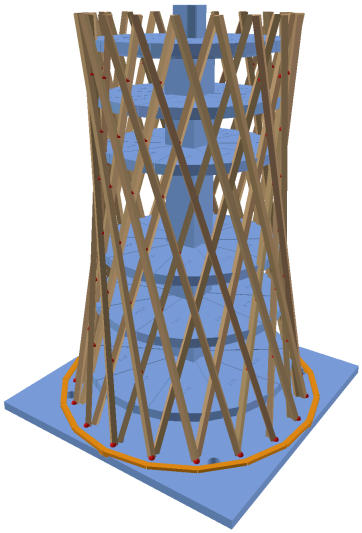
Common mistakes:

- Gluing a crossing before the strip is fully seated in its pocket locks in a crooked strip.
- Coating only one face with balsa cement: it soaks in and the joint stays weak.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **13.62 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 4. Base ring



3D model of this step: `steps/step_04_base.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

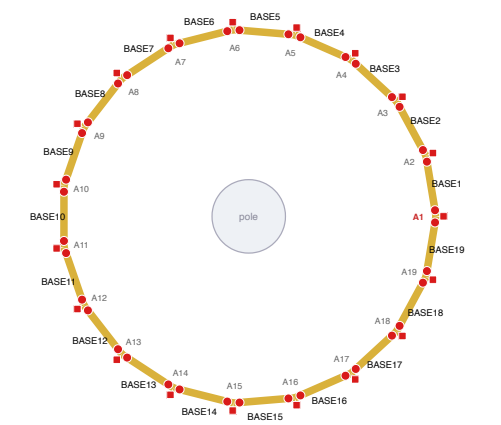
Pieces added: 19 pieces: BASE1, BASE2, BASE3, BASE4, BASE5, BASE6, BASE7, BASE8, BASE9, BASE10, BASE11, BASE12, BASE13, BASE14, BASE15, BASE16, BASE17, BASE18, BASE19

Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

Glue in this step:

- 19 × scarf with **Balsa cement**: apply ≈ 42 mg each, dries to ≤ 13 mg
- 38 × scarf end with **Balsa cement**: apply ≈ 13 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; A1 (red) is at the base-plate mark. Dc

Every joint in this ring (**57 × Balsa cement**):

Piece	End 1 (on strip)	End 2 (on strip)	Scarf overlaps
BASE1	over **A1**: ● Balsa cement	over **A2**: ● Balsa cement	with **BASE2**: ■ Balsa cement, with **BASE19**: ■ Balsa cement
BASE2	over **A2**: ● Balsa cement	over **A3**: ● Balsa cement	with **BASE1**: ■ Balsa cement, with **BASE3**: ■ Balsa cement
BASE3	over **A3**: ● Balsa cement	over **A4**: ● Balsa cement	with **BASE2**: ■ Balsa cement, with **BASE4**: ■ Balsa cement
BASE4	over **A4**: ● Balsa cement	over **A5**: ● Balsa cement	with **BASE3**: ■ Balsa cement, with **BASE5**: ■ Balsa cement
BASE5	over **A5**: ● Balsa cement	over **A6**: ● Balsa cement	with **BASE4**: ■ Balsa cement, with **BASE6**: ■ Balsa cement
BASE6	over **A6**: ● Balsa cement	over **A7**: ● Balsa cement	with **BASE5**: ■ Balsa cement, with **BASE7**: ■ Balsa cement
BASE7	over **A7**: ● Balsa cement	over **A8**: ● Balsa cement	with **BASE6**: ■ Balsa cement, with **BASE8**: ■ Balsa cement
BASE8	over **A8**: ● Balsa cement	over **A9**: ● Balsa cement	with **BASE7**: ■ Balsa cement, with **BASE9**: ■ Balsa cement
BASE9	over **A9**: ● Balsa cement	over **A10**: ● Balsa cement	with **BASE8**: ■ Balsa cement, with **BASE10**: ■ Balsa cement
BASE10	over **A10**: ● Balsa cement	over **A11**: ● Balsa cement	with **BASE9**: ■ Balsa cement, with **BASE11**: ■ Balsa cement
BASE11	over **A11**: ● Balsa cement	over **A12**: ● Balsa cement	with **BASE10**: ■ Balsa cement, with **BASE12**: ■ Balsa cement
BASE12	over **A12**: ● Balsa cement	over **A13**: ● Balsa cement	with **BASE11**: ■ Balsa cement, with **BASE13**: ■ Balsa cement
BASE13	over **A13**: ● Balsa cement	over **A14**: ● Balsa cement	with **BASE12**: ■ Balsa cement, with **BASE14**: ■ Balsa cement
BASE14	over **A14**: ● Balsa cement	over **A15**: ● Balsa cement	with **BASE13**: ■ Balsa cement, with **BASE15**: ■ Balsa cement
BASE15	over **A15**: ● Balsa cement	over **A16**: ● Balsa cement	with **BASE14**: ■ Balsa cement, with **BASE16**: ■ Balsa cement
BASE16	over **A16**: ● Balsa cement	over **A17**: ● Balsa cement	with **BASE15**: ■ Balsa cement, with **BASE17**: ■ Balsa cement
BASE17	over **A17**: ● Balsa cement	over **A18**: ● Balsa cement	with **BASE16**: ■ Balsa cement, with **BASE18**: ■ Balsa cement
BASE18	over **A18**: ● Balsa cement	over **A19**: ● Balsa cement	with **BASE17**: ■ Balsa cement, with **BASE19**: ■ Balsa cement
BASE19	over **A19**: ● Balsa cement	over **A1**: ● Balsa cement	with **BASE18**: ■ Balsa cement, with **BASE1**: ■ Balsa cement

Do this:

1. Take the base-ring pieces (BASE1–BASE19). Each one runs from one foot to the next.
2. Lay a piece on the outside of the B strips so it rests in the shallow groove in the base plate. The groove holds it 0.5 mm above the table, so it never touches the jig.
3. Neighbouring pieces overlap on their 9 mm scarf, centred on a crossing. Hold each piece with its **O** face outwards and its **end 2** (the end where you can see the sloping scarf face) to the right. Work to the right: each new piece's end 1 lies on top of the last piece's end 2. Glue the strip patch and the sloping faces. Glue each end on its strip (dot colour = glue, see the picture). Start at the **A1 B1** mark. Leave the **first** piece's end 1 unglued until the end; then slide the last piece's end 2 underneath it and glue both together.

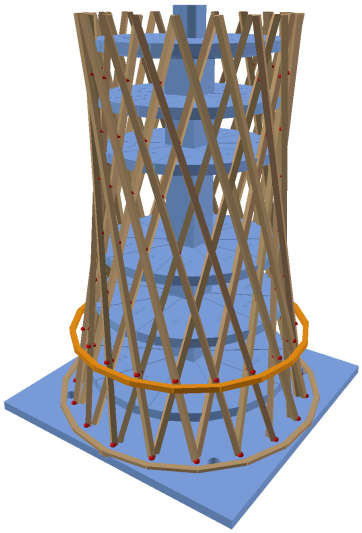
Check:

- □ The base ring sits in its groove all the way round; every piece touches both strips.

Wait at least 1 hour before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **14.85 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 5. Ring 1: 43 mm up



3D model of this step: `steps/step_05_ring_1.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

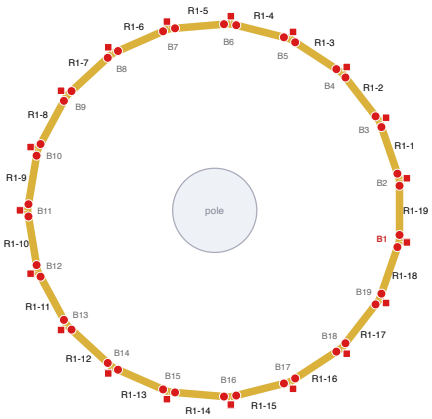
Pieces added: 19 pieces: R1-1, R1-2, R1-3, R1-4, R1-5, R1-6, R1-7, R1-8, R1-9, R1-10, R1-11, R1-12, R1-13, R1-14, R1-15, R1-16, R1-17, R1-18, R1-19

Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

Glue in this step:

- 19 × scarf with **Balsa cement**: apply ≈ 42 mg each, dries to ≤ 13 mg
- 38 × scarf end with **Balsa cement**: apply ≈ 13 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (57 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)	Scarf overlaps
R1-1	over **B2**: ● Balsa cement	over **B3**: ● Balsa cement	with **R1-2**: ■ Balsa cement, with **R1-19**: ■ Balsa cement
R1-2	over **B3**: ● Balsa cement	over **B4**: ● Balsa cement	with **R1-1**: ■ Balsa cement, with **R1-3**: ■ Balsa cement
R1-3	over **B4**: ● Balsa cement	over **B5**: ● Balsa cement	with **R1-2**: ■ Balsa cement, with **R1-4**: ■ Balsa cement
R1-4	over **B5**: ● Balsa cement	over **B6**: ● Balsa cement	with **R1-3**: ■ Balsa cement, with **R1-5**: ■ Balsa cement
R1-5	over **B6**: ● Balsa cement	over **B7**: ● Balsa cement	with **R1-4**: ■ Balsa cement, with **R1-6**: ■ Balsa cement
R1-6	over **B7**: ● Balsa cement	over **B8**: ● Balsa cement	with **R1-5**: ■ Balsa cement, with **R1-7**: ■ Balsa cement
R1-7	over **B8**: ● Balsa cement	over **B9**: ● Balsa cement	with **R1-6**: ■ Balsa cement, with **R1-8**: ■ Balsa cement
R1-8	over **B9**: ● Balsa cement	over **B10**: ● Balsa cement	with **R1-7**: ■ Balsa cement, with **R1-9**: ■ Balsa cement
R1-9	over **B10**: ● Balsa cement	over **B11**: ● Balsa cement	with **R1-8**: ■ Balsa cement, with **R1-10**: ■ Balsa cement
R1-10	over **B11**: ● Balsa cement	over **B12**: ● Balsa cement	with **R1-9**: ■ Balsa cement, with **R1-11**: ■ Balsa cement
R1-11	over **B12**: ● Balsa cement	over **B13**: ● Balsa cement	with **R1-10**: ■ Balsa cement, with **R1-12**: ■ Balsa cement
R1-12	over **B13**: ● Balsa cement	over **B14**: ● Balsa cement	with **R1-11**: ■ Balsa cement, with **R1-13**: ■ Balsa cement
R1-13	over **B14**: ● Balsa cement	over **B15**: ● Balsa cement	with **R1-12**: ■ Balsa cement, with **R1-14**: ■ Balsa cement
R1-14	over **B15**: ● Balsa cement	over **B16**: ● Balsa cement	with **R1-13**: ■ Balsa cement, with **R1-15**: ■ Balsa cement
R1-15	over **B16**: ● Balsa cement	over **B17**: ● Balsa cement	with **R1-14**: ■ Balsa cement, with **R1-16**: ■ Balsa cement
R1-16	over **B17**: ● Balsa cement	over **B18**: ● Balsa cement	with **R1-15**: ■ Balsa cement, with **R1-17**: ■ Balsa cement
R1-17	over **B18**: ● Balsa cement	over **B19**: ● Balsa cement	with **R1-16**: ■ Balsa cement, with **R1-18**: ■ Balsa cement
R1-18	over **B19**: ● Balsa cement	over **B1**: ● Balsa cement	with **R1-17**: ■ Balsa cement, with **R1-19**: ■ Balsa cement
R1-19	over **B1**: ● Balsa cement	over **B2**: ● Balsa cement	with **R1-18**: ■ Balsa cement, with **R1-1**: ■ Balsa cement

Do this:

1. Take the Ring 1 pieces (R1-1–R1-19). This ring goes at crossing level 1, **43 mm** above the table, where the strips' **1R** pencil ticks are.
2. Each piece runs from one crossing to the next, lying on the outside of the B strips. Neighbouring pieces overlap on their 9 mm scarf, centred on a crossing. Hold each piece with its **O** face outwards and its **end 2** (the end where you can see the sloping scarf face) to the right. Work to the right: each new piece's end 1 lies on top of the last piece's end 2. Glue the strip patch and the sloping faces.
3. Centre each scarf on the crossing (the tick). You don't need to measure heights: the crossings set them.
4. Start at the strips above the **A1 B1** mark and go round to the right using the glue map and table on this page. Leave the **first** piece's end 1 unglued until the end; then slide the last piece's end 2 underneath it and glue both together.

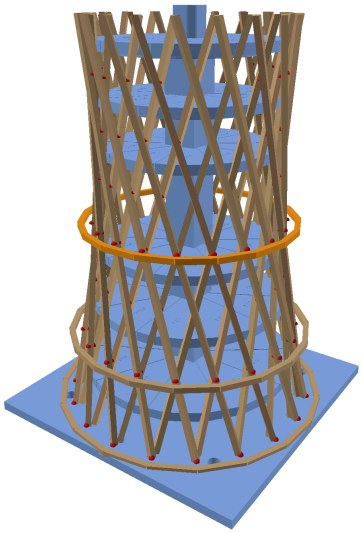
Check:

- □ The ring is level; every piece sits flat on the crossings.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **16.00 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 6. Ring 3: 105 mm up



3D model of this step: `steps/step_06_ring_3.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

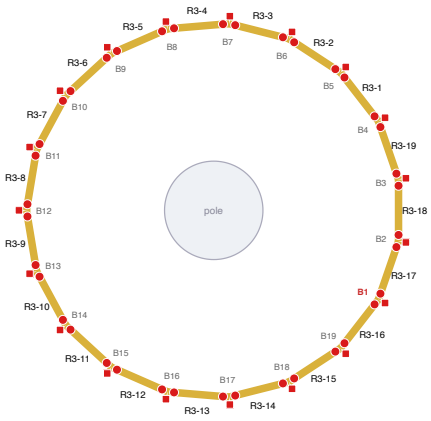
Pieces added: 19 pieces: R3-1, R3-2, R3-3, R3-4, R3-5, R3-6, R3-7, R3-8, R3-9, R3-10, R3-11, R3-12, R3-13, R3-14, R3-15, R3-16, R3-17, R3-18, R3-19

Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

Glue in this step:

- 19 × scarf with **Balsa cement**: apply ≈ 42 mg each, dries to ≤ 13 mg
- 38 × scarf end with **Balsa cement**: apply ≈ 13 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (57 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)	Scarf overlaps
R3-1	over B4: ● Balsa cement	over B5: ● Balsa cement	with R3-2: ■ Balsa cement, with R3-19: ■ Balsa cement
R3-2	over B5: ● Balsa cement	over B6: ● Balsa cement	with R3-1: ■ Balsa cement, with R3-3: ■ Balsa cement
R3-3	over B6: ● Balsa cement	over B7: ● Balsa cement	with R3-2: ■ Balsa cement, with R3-4: ■ Balsa cement
R3-4	over B7: ● Balsa cement	over B8: ● Balsa cement	with R3-3: ■ Balsa cement, with R3-5: ■ Balsa cement
R3-5	over B8: ● Balsa cement	over B9: ● Balsa cement	with R3-4: ■ Balsa cement, with R3-6: ■ Balsa cement
R3-6	over B9: ● Balsa cement	over B10: ● Balsa cement	with R3-5: ■ Balsa cement, with R3-7: ■ Balsa cement
R3-7	over B10: ● Balsa cement	over B11: ● Balsa cement	with R3-6: ■ Balsa cement, with R3-8: ■ Balsa cement
R3-8	over B11: ● Balsa cement	over B12: ● Balsa cement	with R3-7: ■ Balsa cement, with R3-9: ■ Balsa cement
R3-9	over B12: ● Balsa cement	over B13: ● Balsa cement	with R3-8: ■ Balsa cement, with R3-10: ■ Balsa cement
R3-10	over B13: ● Balsa cement	over B14: ● Balsa cement	with R3-9: ■ Balsa cement, with R3-11: ■ Balsa cement
R3-11	over B14: ● Balsa cement	over B15: ● Balsa cement	with R3-10: ■ Balsa cement, with R3-12: ■ Balsa cement
R3-12	over B15: ● Balsa cement	over B16: ● Balsa cement	with R3-11: ■ Balsa cement, with R3-13: ■ Balsa cement
R3-13	over B16: ● Balsa cement	over B17: ● Balsa cement	with R3-12: ■ Balsa cement, with R3-14: ■ Balsa cement
R3-14	over B17: ● Balsa cement	over B18: ● Balsa cement	with R3-13: ■ Balsa cement, with R3-15: ■ Balsa cement
R3-15	over B18: ● Balsa cement	over B19: ● Balsa cement	with R3-14: ■ Balsa cement, with R3-16: ■ Balsa cement
R3-16	over B19: ● Balsa cement	over B1: ● Balsa cement	with R3-15: ■ Balsa cement, with R3-17: ■ Balsa cement
R3-17	over B1: ● Balsa cement	over B2: ● Balsa cement	with R3-16: ■ Balsa cement, with R3-18: ■ Balsa cement
R3-18	over B2: ● Balsa cement	over B3: ● Balsa cement	with R3-17: ■ Balsa cement, with R3-19: ■ Balsa cement
R3-19	over B3: ● Balsa cement	over B4: ● Balsa cement	with R3-18: ■ Balsa cement, with R3-1: ■ Balsa cement

Do this:

1. Take the Ring 3 pieces (R3-1–R3-19). This ring goes at crossing level 3, **105 mm** above the table, where the strips' **3R** pencil ticks are.
2. Each piece runs from one crossing to the next, lying on the outside of the B strips. Neighbouring pieces overlap on their 9 mm scarf, centred on a crossing. Hold each piece with its **O** face outwards and its **end 2** (the end where you can see the sloping scarf face) to the right. Work to the right: each new piece's end 1 lies on top of the last piece's end 2. Glue the strip patch and the sloping faces.
3. Centre each scarf on the crossing (the tick). You don't need to measure heights: the crossings set them.
4. Start at the strips above the **A1 B1** mark and go round to the right using the glue map and table on this page. Leave the **first** piece's end 1 unglued until the end; then slide the last piece's end 2 underneath it and glue both together.

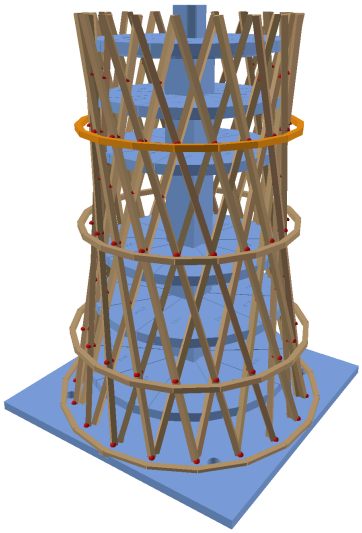
Check:

- □ The ring is level; every piece sits flat on the crossings.

⏸ **Wait at least 1 hour** before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **17.08 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 7. Ring 5: 157 mm up



3D model of this step: `steps/step_07_ring_5.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

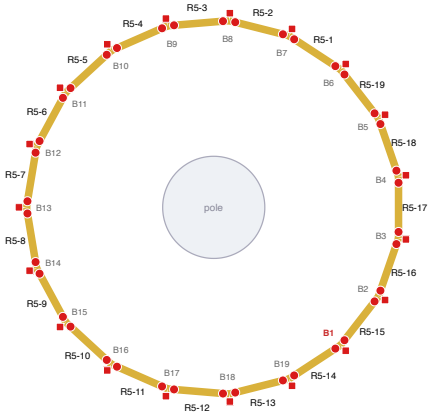
Pieces added: 19 pieces: R5-1, R5-2, R5-3, R5-4, R5-5, R5-6, R5-7, R5-8, R5-9, R5-10, R5-11, R5-12, R5-13, R5-14, R5-15, R5-16, R5-17, R5-18, R5-19

Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

Glue in this step:

- 19 × scarf with **Balsa cement**: apply ≈ 42 mg each, dries to ≤ 13 mg
- 38 × scarf end with **Balsa cement**: apply ≈ 13 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; B1 (red) is at the base-plate mark. Dc

Every joint in this ring (57 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)	Scarf overlaps
R5-1	over B6: ● Balsa cement	over B7: ● Balsa cement	with R5-2: ■ Balsa cement, with R5-19: ■ Balsa cement
R5-2	over B7: ● Balsa cement	over B8: ● Balsa cement	with R5-1: ■ Balsa cement, with R5-3: ■ Balsa cement
R5-3	over B8: ● Balsa cement	over B9: ● Balsa cement	with R5-2: ■ Balsa cement, with R5-4: ■ Balsa cement
R5-4	over B9: ● Balsa cement	over B10: ● Balsa cement	with R5-3: ■ Balsa cement, with R5-5: ■ Balsa cement
R5-5	over B10: ● Balsa cement	over B11: ● Balsa cement	with R5-4: ■ Balsa cement, with R5-6: ■ Balsa cement
R5-6	over B11: ● Balsa cement	over B12: ● Balsa cement	with R5-5: ■ Balsa cement, with R5-7: ■ Balsa cement
R5-7	over B12: ● Balsa cement	over B13: ● Balsa cement	with R5-6: ■ Balsa cement, with R5-8: ■ Balsa cement
R5-8	over B13: ● Balsa cement	over B14: ● Balsa cement	with R5-7: ■ Balsa cement, with R5-9: ■ Balsa cement
R5-9	over B14: ● Balsa cement	over B15: ● Balsa cement	with R5-8: ■ Balsa cement, with R5-10: ■ Balsa cement
R5-10	over B15: ● Balsa cement	over B16: ● Balsa cement	with R5-9: ■ Balsa cement, with R5-11: ■ Balsa cement
R5-11	over B16: ● Balsa cement	over B17: ● Balsa cement	with R5-10: ■ Balsa cement, with R5-12: ■ Balsa cement
R5-12	over B17: ● Balsa cement	over B18: ● Balsa cement	with R5-11: ■ Balsa cement, with R5-13: ■ Balsa cement
R5-13	over B18: ● Balsa cement	over B19: ● Balsa cement	with R5-12: ■ Balsa cement, with R5-14: ■ Balsa cement
R5-14	over B19: ● Balsa cement	over B1: ● Balsa cement	with R5-13: ■ Balsa cement, with R5-15: ■ Balsa cement
R5-15	over B1: ● Balsa cement	over B2: ● Balsa cement	with R5-14: ■ Balsa cement, with R5-16: ■ Balsa cement
R5-16	over B2: ● Balsa cement	over B3: ● Balsa cement	with R5-15: ■ Balsa cement, with R5-17: ■ Balsa cement
R5-17	over B3: ● Balsa cement	over B4: ● Balsa cement	with R5-16: ■ Balsa cement, with R5-18: ■ Balsa cement
R5-18	over B4: ● Balsa cement	over B5: ● Balsa cement	with R5-17: ■ Balsa cement, with R5-19: ■ Balsa cement
R5-19	over B5: ● Balsa cement	over B6: ● Balsa cement	with R5-18: ■ Balsa cement, with R5-1: ■ Balsa cement

Do this:

1. Take the Ring 5 pieces (R5-1–R5-19). This ring goes at crossing level 5, **157 mm** above the table, where the strips' **5R** pencil ticks are.
2. Each piece runs from one crossing to the next, lying on the outside of the B strips. Neighbouring pieces overlap on their 9 mm scarf, centred on a crossing. Hold each piece with its **O** face outwards and its **end 2** (the end where you can see the sloping scarf face) to the right. Work to the right: each new piece's end 1 lies on top of the last piece's end 2. Glue the strip patch and the sloping faces.
3. Centre each scarf on the crossing (the tick). You don't need to measure heights: the crossings set them.
4. Start at the strips above the **A1 B1** mark and go round to the right using the glue map and table on this page. Leave the **first** piece's end 1 unglued until the end; then slide the last piece's end 2 underneath it and glue both together.

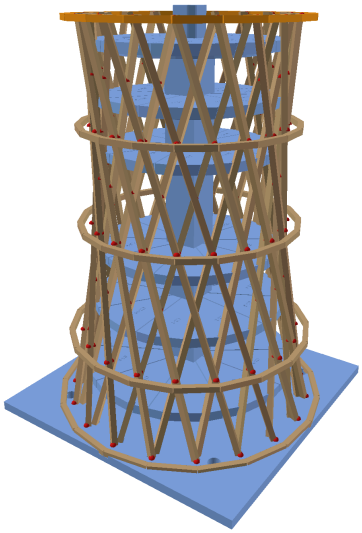
Check:

- □ The ring is level; every piece sits flat on the crossings.

Wait at least 1 hour before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **18.15 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 8. Top ring



3D model of this step: `steps/step_08_cap.step` (open in Onshape: blue = jig, tan = already built, orange = this step, coloured dots = glue)

Jig files on the bench:
the same as the step before: nothing to add or remove

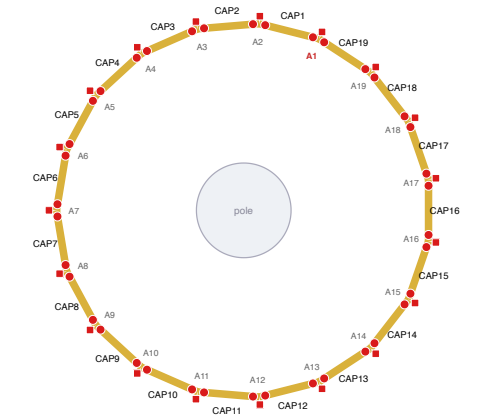
Pieces added: 19 pieces: CAP1, CAP2, CAP3, CAP4, CAP5, CAP6, CAP7, CAP8, CAP9, CAP10, CAP11, CAP12, CAP13, CAP14, CAP15, CAP16, CAP17, CAP18, CAP19

Glue dots: ● **Balsa cement** On the ring maps: ● dot = ring end glued to a strip, ■ square = scarf overlap glued to the next ring piece. Same colour = same glue.

Glue in this step:

- 19 × scarf with **Balsa cement**: apply ≈ 42 mg each, dries to ≤ 13 mg
- 38 × scarf end with **Balsa cement**: apply ≈ 13 mg each, dries to ≤ 4 mg

Glue map of this ring (top view)



Seen from above. The strip each end is glued to is named inside the ring; A1 (red) is at the base-plate mark. Do

Every joint in this ring (57 × Balsa cement):

Piece	End 1 (on strip)	End 2 (on strip)	Scarf overlaps
CAP1	over **A1**: ● Balsa cement	over **A2**: ● Balsa cement	with **CAP2**: ■ Balsa cement, with **CAP19**: ■ Balsa cement
CAP2	over **A2**: ● Balsa cement	over **A3**: ● Balsa cement	with **CAP1**: ■ Balsa cement, with **CAP3**: ■ Balsa cement
CAP3	over **A3**: ● Balsa cement	over **A4**: ● Balsa cement	with **CAP2**: ■ Balsa cement, with **CAP4**: ■ Balsa cement
CAP4	over **A4**: ● Balsa cement	over **A5**: ● Balsa cement	with **CAP3**: ■ Balsa cement, with **CAP5**: ■ Balsa cement
CAP5	over **A5**: ● Balsa cement	over **A6**: ● Balsa cement	with **CAP4**: ■ Balsa cement, with **CAP6**: ■ Balsa cement
CAP6	over **A6**: ● Balsa cement	over **A7**: ● Balsa cement	with **CAP5**: ■ Balsa cement, with **CAP7**: ■ Balsa cement
CAP7	over **A7**: ● Balsa cement	over **A8**: ● Balsa cement	with **CAP6**: ■ Balsa cement, with **CAP8**: ■ Balsa cement
CAP8	over **A8**: ● Balsa cement	over **A9**: ● Balsa cement	with **CAP7**: ■ Balsa cement, with **CAP9**: ■ Balsa cement
CAP9	over **A9**: ● Balsa cement	over **A10**: ● Balsa cement	with **CAP8**: ■ Balsa cement, with **CAP10**: ■ Balsa cement
CAP10	over **A10**: ● Balsa cement	over **A11**: ● Balsa cement	with **CAP9**: ■ Balsa cement, with **CAP11**: ■ Balsa cement
CAP11	over **A11**: ● Balsa cement	over **A12**: ● Balsa cement	with **CAP10**: ■ Balsa cement, with **CAP12**: ■ Balsa cement
CAP12	over **A12**: ● Balsa cement	over **A13**: ● Balsa cement	with **CAP11**: ■ Balsa cement, with **CAP13**: ■ Balsa cement
CAP13	over **A13**: ● Balsa cement	over **A14**: ● Balsa cement	with **CAP12**: ■ Balsa cement, with **CAP14**: ■ Balsa cement
CAP14	over **A14**: ● Balsa cement	over **A15**: ● Balsa cement	with **CAP13**: ■ Balsa cement, with **CAP15**: ■ Balsa cement
CAP15	over **A15**: ● Balsa cement	over **A16**: ● Balsa cement	with **CAP14**: ■ Balsa cement, with **CAP16**: ■ Balsa cement
CAP16	over **A16**: ● Balsa cement	over **A17**: ● Balsa cement	with **CAP15**: ■ Balsa cement, with **CAP17**: ■ Balsa cement
CAP17	over **A17**: ● Balsa cement	over **A18**: ● Balsa cement	with **CAP16**: ■ Balsa cement, with **CAP18**: ■ Balsa cement
CAP18	over **A18**: ● Balsa cement	over **A19**: ● Balsa cement	with **CAP17**: ■ Balsa cement, with **CAP19**: ■ Balsa cement
CAP19	over **A19**: ● Balsa cement	over **A1**: ● Balsa cement	with **CAP18**: ■ Balsa cement, with **CAP1**: ■ Balsa cement

Do this:

1. Take the top-ring pieces (CAP1–CAP19).
2. Lay each piece on the outside of the B strips, with its top flush with the strip tops. Neighbouring pieces overlap on their 9 mm scarf, centred on a crossing. Hold each piece with its **O** face outwards and its **end 2** (the end where you can see the sloping scarf face) to the right. Work to the right: each new piece's end 1 lies on top of the last piece's end 2. Glue the strip patch and the sloping faces. Glue both ends (dot colour = glue).
3. Start at the strips above the **A1 B1** mark and go round to the right, using the glue map and table on this page. Leave the **first** piece's end 1 unglued until the end; then slide the last piece's end 2 underneath it and glue both together.

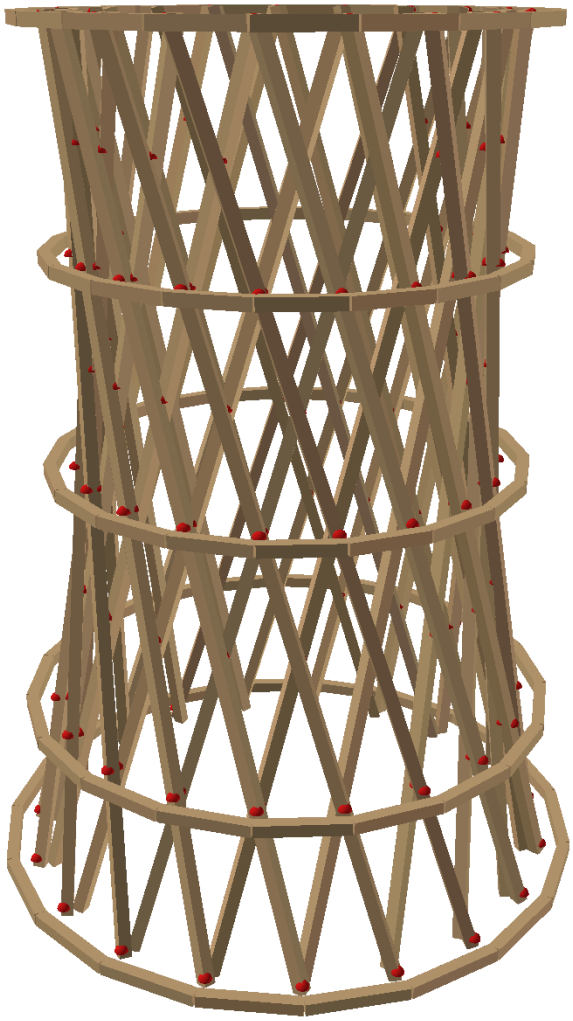
Check:

- □ The top ring is flush with the strip ends all the way round.

Wait at least 1 hour before the next step so the glue can grab. Don't load or remove anything for 12 hours.

⚖ Weigh after this step, once the glue is dry: no more than **19.24 g**. If it is heavier, you are using too much glue: wipe joints and use less on the next step.

Step 9. Cure, remove the jig and finish



Jig files on the bench:
none (the jig comes off in this step)

Pieces added: none

3D model of this step: `steps/step_09_remove.step` (open in Onshape:
blue = jig, tan = already built, orange = this step, coloured dots = glue)

Do this:

1. Leave everything for **24 hours** so every glue reaches full strength.
2. Peel off every masking-tape tab.
3. Lift the tower off the base plate, mast and all: push the feet up gently through the finger holes, a little at a time all round, until they come out of the pockets. Hold the tower by its rings, never by one strip.
4. Hold the tower over a box and pull the mast **M** straight up out of the top.
5. Now take the jig out **from the bottom up**, in the reverse of how you stacked it: slide **SP1** down and out; then push each **C1** segment about 3 mm inwards off its strips, tip it and lower it out of the bottom; then **SP2**, then the **C2** segments, and so on up to **C6**. Every part comes out through the wide bottom of the tower.
6. Never pull a segment outwards or upwards through the strips. If one sticks, push it inwards firmly or cut it with a knife: it's only a jig part.
7. Check the tower on a flat table. If it rocks, lay the sandpaper board on the table, rest a flat book on the top ring, and slide the tower in short straight strokes (2–3 cm, no twisting) with light pressure. Check after every 5 strokes.
8. Do the same for the top, upside down, so the load plate will bear on every strip at once.
9. Weigh the finished tower.

Check:

- □ It weighs no more than the limit.
- □ It stands upright without rocking.
- □ The pole slides through without touching.

Common mistakes:

- Sanding in circles or pressing hard snaps feet. Short straight strokes, light pressure.

11. Final checks

- □ Mass is no more than **19.24 g** (limit 20 g).
- □ It stands on a flat table without rocking; every foot touches.
- □ The top is flat: lay a ruler across it in three directions and check for gaps.
- □ No joint has a visible gap or loose piece. Press gently on each ring; nothing moves.
- □ The pole slides through without touching (about 31 mm gap).
- □ No masking tape or jig parts left inside.

12. Test day

1. Weigh the tower in front of the tester and record it.
2. Slide it over the pole, centred. Check the gap all round.
3. Seat the load plate gently and check it touches the top ring and strip tops all the way round. Don't shim; sand if needed.
4. Add load smoothly and evenly. Shock loads break brittle joints first.
5. Record the load at failure and where it failed (which piece or joint). Compare it with the prediction (83 kg) and the weakest point listed above.

Tower mass (g)	Failure load (kg)	Where it failed	Efficiency (load ÷ mass)

13. Troubleshooting

Problem	Likely cause	Fix
Strips won't sit in the collar notches	A segment on the wrong number, upside down, or spacers swapped	Check each segment sits on its own number with its ID on top, and each spacer's disc says "under" the collar above it.
A crossing doesn't touch (gap)	Strip not fully seated, or bowed	Re-seat the strip in its notches. Swap bowed strips for straight ones. Never fill a gap with glue.
Over mass at a weigh-in	Too much glue	Use less on the remaining joints; wipe squeeze-out immediately. Do the glue test.
Glue stuck the tower to the jig	Glue ran onto the plastic	Free it with a thin blade; sand the joint smooth. Keep packing tape on the base plate.
Ring not level	Pieces glued at slightly different heights	Measure each piece from the table before the glue sets.
Collar segment won't come out	Glue on the notch	Push it inwards firmly; if stuck, cut the segment with a knife. It is only a jig part.
Tower rocks on the table	Feet not level	Short straight strokes on the sandpaper board with a book on top (Step 9). Never circles.
Balsa cement joint stays soft	Single coat, or not cured	Use the double-glue method and wait 12 h.

Fixing mistakes

What went wrong	What to do
A strip breaks before it's glued	Cut a new one from a spare stick in channel 1, letter it, mark it in the marking gauge and sand its ends, exactly like the others.
A glued piece breaks, or a joint is glued in the wrong place	Balsa cement softens with acetone: dab a little on the joint with a cotton bud, wait a minute, and ease the pieces apart. Scrape off the old glue, let it dry, and glue again. Replace a broken piece with a spare.
A scarf sanded too far (the piece is now too short to overlap)	Use one of the spare ring pieces.
A strip end sanded too far (strip short of the END notch)	Over 1 mm short: cut a new strip. The tops must be level for the load plate.
You run out of sticks	You have 2 spare; the offcuts at the end of the strip sticks are long enough for ring pieces.
A jig part is lost or broken	Reprint just that file; its name is in the table in section 8 and Appendix B.
Over mass at the end	Don't sand pieces thinner. Look for big glue blobs and scrape them off with a blade.

Appendix A: every piece

ID	Group	Length mm	End 1	End 2	Glue at its joints	Max mg
A1	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A2	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A3	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A4	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A5	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A6	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A7	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A8	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A9	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A10	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A11	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A12	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A13	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A14	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A15	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A16	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A17	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A18	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
A19	Strip A	220.2	level cut, ~17° off square	level cut, ~17° off square	10× Balsa cement	327
B1	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B2	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B3	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B4	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B5	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B6	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B7	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B8	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B9	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B10	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B11	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B12	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B13	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B14	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B15	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B16	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B17	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B18	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
B19	Strip B	220.2	level cut, ~17° off square	level cut, ~17° off square	12× Balsa cement	327
BASE1	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE2	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE3	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE4	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE5	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE6	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE7	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE8	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE9	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE10	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE11	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE12	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE13	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE14	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44

BASE15	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE16	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE17	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE18	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
BASE19	Base ring	30.0	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	44
CAP1	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP2	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP3	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP4	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP5	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP6	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP7	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP8	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP9	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP10	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP11	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP12	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP13	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP14	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP15	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP16	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP17	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP18	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
CAP19	Top cap	25.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R1-1	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-2	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-3	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-4	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-5	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-6	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-7	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-8	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-9	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-10	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-11	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-12	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-13	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-14	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-15	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-16	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-17	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-18	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R1-19	Ring 1	27.3	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	41
R3-1	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-2	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-3	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-4	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-5	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-6	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-7	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-8	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-9	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-10	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-11	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-12	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37

R3-13	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-14	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-15	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-16	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-17	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-18	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R3-19	Ring 3	24.6	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	37
R5-1	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-2	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-3	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-4	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-5	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-6	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-7	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-8	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-9	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-10	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-11	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-12	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-13	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-14	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-15	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-16	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-17	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-18	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36
R5-19	Ring 5	24.1	scarf: 1-in-3 slope through the thickness, ~9 mm	scarf: 1-in-3 slope through the thickness, ~9 mm	4× Balsa cement	36

Appendix B: jig and model files

Jig files (`jig/` ; each has a .stl to print and a .step for Onshape) and the steps that use them:

File	Used in step
`base_plate.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_1_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_2_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_3_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_4_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_5_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8

`collar_5_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg7.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg8.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg9.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`collar_6_seg10.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`cutting_guide_1.stl / .step`	Step 0 (preparing parts)
`end_sanding_jig_17deg.stl / .step`	Step 0 (preparing parts)
`fit_test_coupon.stl / .step`	Step 0 (preparing parts)
`marking_gauge.stl / .step`	Step 0 (preparing parts)
`mast.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`scarf_sanding_guide.stl / .step`	Step 0 (preparing parts)
`spacer_1.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`spacer_2.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`spacer_3.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`spacer_4.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`spacer_5.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8
`spacer_6.stl / .step`	1, 2, 3, 4, 5, 6, 7, 8

Step models (`steps/` , open in Onshape):

Step	What	File
Step 1	Set up the jig	`steps/step_01_jig.step`
Step 2	Inner strips (set A)	`steps/step_02_strips_a.step`
Step 3	Outer strips (set B) and the crossings	`steps/step_03_strips_b.step`
Step 4	Base ring	`steps/step_04_base.step`
Step 5	Ring 1: 43 mm up	`steps/step_05_ring_1.step`
Step 6	Ring 3: 105 mm up	`steps/step_06_ring_3.step`
Step 7	Ring 5: 157 mm up	`steps/step_07_ring_5.step`
Step 8	Top ring	`steps/step_08_cap.step`
Step 9	Cure, remove the jig and finish	`steps/step_09_remove.step`

Whole tower: `tower.step` . **Assembled jig:** `jig/jig_assembly.step` .

Importing into Onshape: in a document click **+** → **Import…** and choose the file. STEP files come in as proper editable parts; STL files come in as meshes (fine for viewing, awkward to edit).